6.2.3.1

Sunday, April 26, 2026 4:29 PM

Exercise 102. Consider the Rank-1 Constraint System for points on the Tiny-jubjub curve from example 121. Compute an associated QAP for this R1CS and double check your computation using sage.

Example 121 (R1CS for the points of the Tiny-jubjub curve). Consider the languages $L_{tiny.jj.1}$ from example 119, which consists of words $< I_1, I_2 >$ over the alphabet $\mathbb{F}_{13}$ such that $3 \cdot I_1^2 + I_2^2 = 1 + 8 \cdot I_1^2 \cdot I_2^2$.

We derive a Rank-1 Constraint System such that its solutions are in a 1:1 correspondence with words in $L_{tiny.jj.1}$. To achieve this, we first rewrite the defining equation:

$$\begin{aligned} 3 \cdot I_1^2 + I_2^2 &= 1 + 8 \cdot I_1^2 \cdot I_2^2 && \Leftrightarrow \\ 0 &= 1 + 8 \cdot I_1^2 \cdot I_2^2 - 3 \cdot I_1^2 - I_2^2 && \Leftrightarrow \\ 0 &= 1 + 8 \cdot I_1^2 \cdot I_2^2 + 10 \cdot I_1^2 + 12 \cdot I_2^2 \end{aligned}$$

Since R1CSs are systems of quadratic equations, we have to reformulate this expression into a system of quadratic equations. To do so, we have to introduce new variables that constrain intermediate steps in the computation, and we have to decide if those variables should be instance or witness variables. We decide to declare all new variables as witness variables, and get the following constraints:

$$\begin{aligned} I_1 \cdot I_1 &= W_1 && \text{constraint 1} \\ I_2 \cdot I_2 &= W_2 && \text{constraint 2} \\ (8 \cdot W_1) \cdot W_2 &= W_3 && \text{constraint 3} \\ (12 \cdot W_2 + W_3 + 10 \cdot W_1 + 1) \cdot 1 &= 0 && \text{constraint 4} \end{aligned}$$

To see that these four quadratic equations qualify as a Rank-1 Constraint System according to definition 6.2.1.1, choose the parameter $n = 2$, $m = 3$, $k = 4$, and the following values:

$$\begin{array}{llllll} a_0^1 = 0 & a_1^1 = 1 & a_2^1 = 0 & a_3^1 = 0 & a_4^1 = 0 & a_5^1 = 0 \\ a_0^2 = 0 & a_1^2 = 0 & a_2^2 = 1 & a_3^2 = 0 & a_4^2 = 0 & a_5^2 = 0 \\ a_0^3 = 0 & a_1^3 = 0 & a_2^3 = 0 & a_3^3 = 8 & a_4^3 = 0 & a_5^3 = 0 \\ a_0^4 = 1 & a_1^4 = 0 & a_2^4 = 0 & a_3^4 = 10 & a_4^4 = 12 & a_5^4 = 1 \end{array}$$

$$\begin{array}{llllll} b_0^1 = 0 & b_1^1 = 1 & b_2^1 = 0 & b_3^1 = 0 & b_4^1 = 0 & b_5^1 = 0 \\ b_0^2 = 0 & b_1^2 = 0 & b_2^2 = 1 & b_3^2 = 0 & b_4^2 = 0 & b_5^2 = 0 \\ b_0^3 = 0 & b_1^3 = 0 & b_2^3 = 0 & b_3^3 = 0 & b_4^3 = 1 & b_5^3 = 0 \\ b_0^4 = 1 & b_1^4 = 0 & b_2^4 = 0 & b_3^4 = 0 & b_4^4 = 0 & b_5^4 = 0 \end{array}$$

$$\begin{array}{llllll} c_0^1 = 0 & c_1^1 = 0 & c_2^1 = 0 & c_3^1 = 1 & c_4^1 = 0 & c_5^1 = 0 \\ c_0^2 = 0 & c_1^2 = 0 & c_2^2 = 0 & c_3^2 = 0 & c_4^2 = 1 & c_5^2 = 0 \\ c_0^3 = 0 & c_1^3 = 0 & c_2^3 = 0 & c_3^3 = 0 & c_4^3 = 0 & c_5^3 = 1 \\ c_0^4 = 0 & c_1^4 = 0 & c_2^4 = 0 & c_3^4 = 0 & c_4^4 = 0 & c_5^4 = 0 \end{array}$$

With this choice, the Rank-1 Constraint System of our Tiny-jubjub curve point problem can be written in its most general form as follows:

```
(sage) me@LAPTOP-GP63ATSE:~$ sage
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│ SageMath version 10.5, Release Date: 2024-12-04                                                          │
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│ Using Python 3.10.17. Type "help()" for help.                    │
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sage: F13 = GF(13)
sage: F13t.<t> = F13[]
sage: T = F13t((t-1)*(t-2)*(t-3)*(t-4))
sage: T
t^4 + 3*t^3 + 9*t^2 + 2*t + 11
sage: A1 = F13t.lagrange_polynomial([(1,0),(2,0),(3,0),(4,1)])
sage: A1
11*t^3 + 12*t^2 + 4*t + 12
sage: A2 = F13t.lagrange_polynomial([(1,1),(2,0),(3,0),(4,0)])
sage: A2
2*t^3 + 8*t^2 + 4
sage: A3 = F13t.lagrange_polynomial([(1,0),(2,1),(3,0),(4,0)])
sage: A3
7*t^3 + 9*t^2 + 3*t + 7
sage: A4 = F13t.lagrange_polynomial([(1,0),(2,0),(3,8),(4,10)])
sage: A4
2*t^3 + 5*t^2 + 10*t + 9
sage: A5 = F13t.lagrange_polynomial([(1,0),(2,0),(3,0),(4,12)])
sage: A5
2*t^3 + t^2 + 9*t + 1
sage: A6 = F13t.lagrange_polynomial([(1,0),(2,0),(3,0),(4,1)])
sage: A6
11*t^3 + 12*t^2 + 4*t + 12
sage: B1 = F13t.lagrange_polynomial([(1,0),(2,0),(3,0),(4,1)])
sage: B1
11*t^3 + 12*t^2 + 4*t + 12
sage: B2 = F13t.lagrange_polynomial([(1,1),(2,0),(3,0),(4,0)])
sage: B2
2*t^3 + 8*t^2 + 4
sage: B3 = F13t.lagrange_polynomial([(1,0),(2,1),(3,0),(4,0)])
sage: B3
7*t^3 + 9*t^2 + 3*t + 7
sage: B4 = F13t.lagrange_polynomial([(1,0),(2,0),(3,0),(4,0)])
sage: B4
0
sage: B5 = F13t.lagrange_polynomial([(1,0),(2,0),(3,1),(4,0)])
sage: B5
6*t^3 + 10*t^2 + 6*t + 4
sage: C4 = F13t.lagrange_polynomial([(1,1),(2,0),(3,0),(4,0)])
sage: C4
2*t^3 + 8*t^2 + 4
sage: C5 = F13t.lagrange_polynomial([(1,0),(2,1),(3,0),(4,0)])
sage: C5
7*t^3 + 9*t^2 + 3*t + 7
sage: C6 = F13t.lagrange_polynomial([(1,0),(2,0),(3,1),(4,0)])
sage: C6
6*t^3 + 10*t^2 + 6*t + 4

QAP = {
t^4 + 3*t^3 + 9*t^2 + 2*t + 11,
{ 11*t^3 + 12*t^2 + 4*t + 12, 2*t^3 + 8*t^2 + 4, 7*t^3 + 9*t^2 + 3*t + 7, 2*t^3 + 5*t^2 + 10*t + 9, 2*t^3
+ t^2 + 9*t + 1, 11*t^3 + 12*t^2 + 4*t + 12},
{11*t^3 + 12*t^2 + 4*t + 12, 2*t^3 + 8*t^2 + 4, 7*t^3 + 9*t^2 + 3*t + 7, 0, 6*t^3 + 10*t^2 + 6*t + 4,0 },
{0,0,0, 2*t^3 + 8*t^2 + 4, 7*t^3 + 9*t^2 + 3*t + 7, 6*t^3 + 10*t^2 + 6*t + 4},
```

}